

# Feng-Kai (Leonel) Huang

PH.D. STUDENT · NATIONAL TAIWAN UNIVERSITY

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## Summary

Feng-Kai (Leonel) Huang is a Ph.D. student in Computer Science and Information Engineering at National Taiwan University (NTU), where he is a member of the Communications and Multimedia Laboratory ([CMLab](#)) in the Artificial Intelligence and MultiMedia ([AIMM](#)) Research Group. His research is supervised by Prof. [Wen-Huang Cheng](#) and Prof. [Hong-Han Shuai](#). His consistent academic excellence is highlighted by his induction into the Phi Tau Phi Scholastic Honor Society twice during his undergraduate and master's studies.

His research interests are centered on **Artificial Intelligence**, **Machine Learning**, **Computer Vision**, and **Multimedia Processing**. He is passionate about developing innovative algorithms to solve real-world challenges in computer vision, with publications in international conferences such as ACM MM, ICCV and WACV. He is always enthusiastic about exploring new ideas and open to potential research collaborations.

## Education

### NTU (National Taiwan University)

Taipei, Taiwan

PH.D. STUDENT IN COMPUTER SCIENCE AND INFORMATION ENGINEERING

Aug. 2024 - Present

- GPA: 4.26/4.3 (Fall Semester 2024-2025).
- Advisor: Prof. [Wen-Huang Cheng](#) and Prof. [Hong-Han Shuai](#).
- Laboratory: Communications and Multimedia Laboratory ([CMLab](#)) - Artificial Intelligence and Multimedia ([AIMM](#)) Research Group.

### NYCU (National Yang Ming Chiao Tung University)

Hsinchu, Taiwan

MASTER OF THE GRADUATE DEGREE PROGRAM OF ARTIFICIAL INTELLIGENCE

Sep. 2022 - Jul. 2024

- GPA: 4.25/4.3.
- Advisor: Prof. [Hong-Han Shuai](#).
- Laboratory: Big data Analytics and Social Intelligent Computing ([BASIC](#)) Laboratory.

### CYCU (Chung Yuan Christian University)

Taoyuan, Taiwan

BACHELOR OF ELECTRICAL ENGINEERING

Sep. 2018 - Jun. 2022

- GPA: 3.98/4.
- Rank: 1/104 (ranked in department).

## Publications

### CONFERENCE

- **F.-K. Huang**, B.-L. Huang, L.-W. Tsao, J.-C. Wu, H.-H. Shuai, and W.-H. Cheng. "Flowing Crowd to Count Flows: A Self-Supervised Framework for Video Individual Counting". In: *ACM International Conference on Multimedia (MM)*. 2025. (Regular Track Paper) [\[Website\]](#) [\[Code\]](#)
- **F.-K. Huang**, H.-W. Xu, C.-C. Lee, H.-Y. Tu, H.-H. Shuai, and W.-H. Cheng. "OinkTrack: An Ultra-Long-Term Dataset for Multi-Object Tracking and Re-Identification of Group-Housed Pigs". In: *ACM International Conference on Multimedia (MM)*. 2025. (Dataset Track Paper) [\[Website\]](#) [\[Code\]](#)
- B.-L. Huang, Z.-X. Ni, **F.-K. Huang**, H.-H. Shuai, and W.-H. Cheng. "When Anchors Meet Cold Diffusion: A Multi-Stage Approach to Lane Detection". In: *International Conference on Computer Vision (ICCV)*. 2025. (Accepting rate = 24% (2698/11239))
- C.-Y. Chiu, **F.-K. Huang**, T.-F. Hsiao, H.-H. Shuai, and W.-H. Cheng. "IterDiff: Training-Free Iterative Face Editing via Efficient CLIP-guided Memory Bank". In: *IEEE International Conference on Image Processing (ICIP)*. 2025. [\[Paper\]](#) [\[Code\]](#)
- C.-L. Wan, **F.-K. Huang**, and H.-H. Shuai. "Density-Based Flow Mask Integration via Deformable Convolution for Video People Flux Estimation". In: *IEEE Winter Conference on Application of Computer Vision (WACV)*. 2024. [\[Paper\]](#) [\[Code\]](#)
- Y.-P. Liao, **F.-K. Huang**, Y.-J. Xia, and H. Cheng. "Smart Speaker Based on Detection of Millimeter Wave". In: *IEEE International Conference on Consumer Electronics-Taiwan (ICCE-TW)*. 2022. [\[Paper\]](#)

### UNDER REVIEW/IN PREPARATION

- **F.-K. Huang**, K.-J. Weng, J.-C. Wu, H.-H. Shuai, and W.-H. Cheng. "Density-Assisted Adaptive Alignment Network for Video Individual Counting". In: *Submitted to ACM Transactions on Intelligent Systems and Technology (TIST)* (2025)

## Honors & Awards

### ACADEMIC HONORS

|      |                                                                                                                                                                                                                      |      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 2024 | <b>Graduate Research Fellowship</b> , National Science and Technology Council <a href="#">[Link]</a>                                                                                                                 | NTU  |
| 2024 | <b>Honorary Member</b> , The Phi Tau Phi Scholastic Honor Society of the Republic of China <a href="#">[Link]</a>                                                                                                    | NYCU |
| 2022 | <b>Certificate of Excellence</b> , Spring Semester 2022-2023 <a href="#">[Link]</a>                                                                                                                                  | NYCU |
| 2022 | <b>Honorary Member</b> , The Phi Tau Phi Scholastic Honor Society of the Republic of China <a href="#">[Link]</a>                                                                                                    | CYCU |
| 2018 | <b>Certificate of Excellence</b> , Fall & Spring Semesters 2018-2019, Fall & Spring Semesters 2019-2020, Fall & Spring Semesters 2020-2021 <a href="#">[Link]</a>                                                    | CYCU |
| 2018 | <b>Subjects Merit Awards</b> , Calculus I, Circuit Theory I & II, Engineering Mathematics II, Electronics II, The Principles of Communications, Business Data Communication, Electromagnetism <a href="#">[Link]</a> | CYCU |

### INTERNATIONAL AWARDS

|      |                                                                                                                            |        |
|------|----------------------------------------------------------------------------------------------------------------------------|--------|
| 2021 | <b>Gold Prize</b> , 2021 Seoul International Invention Fair   Korea Invention Promotion Association <a href="#">[Link]</a> | Online |
|------|----------------------------------------------------------------------------------------------------------------------------|--------|

### DOMESTIC AWARDS

|      |                                                                                                                                                                                                                 |                |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2022 | <b>Honorable Mention Award</b> , 2022 AI CUP Competition on UAV Smart Counting   Ministry of Education <a href="#">[Link]</a>                                                                                   | Taipei, Taiwan |
| 2021 | <b>Honorable Award</b> , 2021 Student Engineering Papers Competition   Chinese Institute of Engineers <a href="#">[Link]</a>                                                                                    | Online         |
| 2021 | <b>1<sup>st</sup> Place (TQC+)</b> , International ICT Innovative Services Awards 2021   Ministry of Economic Affairs, Ministry of Education, Chinese Society for Information Management <a href="#">[Link]</a> | Online         |
| 2021 | <b>3<sup>rd</sup> Place (IP)</b> , International ICT Innovative Services Awards 2021   Ministry of Economic Affairs, Ministry of Education, Chinese Society for Information Management <a href="#">[Link]</a>   | Online         |
| 2020 | <b>3<sup>rd</sup> Place (TQC+)</b> , International ICT Innovative Services Awards 2020   Ministry of Economic Affairs, Ministry of Education, Chinese Society for Information Management <a href="#">[Link]</a> | Taipei, Taiwan |

## Work Experience

### National Taiwan University

Taipei, Taiwan

TEACHING ASSISTANT – ARTIFICIAL INTELLIGENCE (SPRING SEMESTER 2024-2025)

Mar. 2025 - Jun. 2025

- Co-designed three hands-on assignments covering MLLM, RAG, and Multi-Agent Systems, plus an open-ended final project.
- Refined each assignment through self-testing to ensure accessibility and challenge for students of all levels.
- Managed all assessment activities.

## Patents

- Y.-P. Liao, **F.-K. Huang**, Y.-J. Xia, and H. Cheng. “Rotate-type smart speaker and rotate-type home appliance controller”. TWI817194B. C. Y. C. University. Oct. 1, 2023 [\[Link\]](#)
- F.-K. Huang**, Y.-J. Xia, and H. Cheng. “System for detecing fecal and toilet lid with fecal detection thereof”. TWI788834B. C. Y. C. University. Jan. 1, 2023 [\[Link\]](#)

## Skills

|                          |                                                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------------|
| <b>Programming</b>       | Python, C/C++, LaTeX, Verilog                                                                        |
| <b>Research Interest</b> | Artificial Intelligence, Machine Learning, Computer Vision, Multimedia Processing, Generative Models |
| <b>Developer Tools</b>   | Jetson Nano, Raspberry Pi, Arduino                                                                   |
| <b>Languages</b>         | Mandarin (native), English ( <b>TOEIC</b> 800/990)                                                   |